

Solution to Hedva 1M moed bet exam of 6/3/2000

March 13, 2000

Part A

1. Correct answer: $f(x) \leq \frac{1}{x}$ for all $x \geq 1$.
2. Correct answer: 3.
3. Correct answer: $\frac{\pi}{2} - \frac{1}{2}\ln(2)$.
4. Correct answer: $\{|S_n|\}_{n=1}^{\infty}$ is a bounded sequence.
5. Correct answer: $\lim_{n \rightarrow \infty} \frac{b_n + 1}{a_n - 2}$ exists and is finite.

Part B

NOT TRUE: If $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are monotone sequences, then $\{a_n + b_n\}_{n=1}^{\infty}$ converges in the large sense.

TRUE: $\lim_{x \rightarrow 0} \frac{\ln(1 + \sin(x^3))}{x^2} = 0$.

TRUE: If f is continuous in $[0, \infty)$ and $F(x) = \int_0^{x^2} f(t)dt$ is strictly increasing, then $f(x) \geq 0$ in $[0, \infty)$.

NOT TRUE: $\int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{x^4} dx$ converges.

NOT TRUE: If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$ and $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ are sequences which converge to 1, then $\lim_{n \rightarrow \infty} \frac{f(a_n)}{g(b_n)} = 1$.

NOT TRUE: If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous in $[0, \infty)$ such that $f(0) = 0$ and $\lim_{x \rightarrow \infty} f(x)$ exists and is finite, then there exists an $\epsilon > 0$ such that for every $\delta > 0$: if $0 < x < \delta$, then $|f(x)| < \epsilon$.

TRUE: The equation $x^3 + 3x^2 - 5 = 0$ has exactly one solution in $(-\infty, \infty)$.

NOT TRUE: If f is differentiable and f attains every real value, then f' is not bounded.

NOT TRUE: If f is differentiable and f' attains every real value, then f is not bounded.

Part C

15.* $f : [0, \infty) \rightarrow \mathbb{R}$ is integrable in every bounded interval such that $\sum_{n=1}^{\infty} f(n)$ converges, but $\int_1^{\infty} f(x)dx$ diverges.

Example: $f(x) = \sin(\pi x)$ satisfies the conditions.

15.** $\lim_{x \rightarrow 3} f(x) = 5$, and the sequence $\{a_n\}_{n=1}^{\infty}$ tends to 3 yet $f(a_n) < 0$.

Answer: This is impossible, because by definition of $\lim$, since

$\lim_{x \rightarrow 3} f(x) = 5$, then for every $\epsilon > 0$ there exists $\delta = \delta(\epsilon) > 0$ such that whenever $0 < |x - 3| < \delta$ then $5 - \epsilon < f(x) < 5 + \epsilon$, and in particular for $\epsilon = 1$ it follows that $f(x) > 4$ whenever $0 < |x - 3| < \delta(1)$. Thus, since $a_n \rightarrow 3$, there exists an integer N such that for all $n > N$ $|a_n - 3| < \delta(1)$ and therefore $f(a_n) > 4$. Therefore it is impossible that $f(a_n) < 0$!

15.*** $\{f_n(x)\}_{n=1}^{\infty}$ is a sequence of continuous functions which converges pointwise to a continuous function $f(x)$ in $[0, 1]$ but does not converge uniformly.

Example: The continuous functions $f_n(x)$ defined to be: $\sin(n\pi x)$ in $0 \leq x \leq 1/n$ and $= 0$ in $1/n \leq x \leq 1$, converge pointwise to the zero function $f(x) \equiv 0$, but not uniformly because if we take $x_n = \pi/2n$ then $x_n \rightarrow 0$ yet $f_n(x_n) = 1$, i.e. even for "large" values n , the functions $f_n(x)$ are "far" from the zero function even for points x which are "very close" to $x = 0$.

Part D (homework problem)

16. Write down the linear approximation of the function $f(x) = x\sqrt{x}$ about the point $x = 1$ and estimate the remainder for $|x - 1| < 0.1$.

Answer: By Taylor's formula with $n = 1$ since $f(x)$ and its derivatives are continuous in the neighborhood of the point $a = 1$

$$f(x) = f(1) + f'(1)(x - 1) + R_1(x) = 1 + \frac{3}{2}(x - 1) + R_1(x),$$

where the remainder $R_1(x) = f''(\xi)(x - 1)^2/2!$ and $|\xi - 1| < |x - 1|$.

Since $|x - 1| < 0.1$, therefore $|\xi - 1| < 0.1$, and so $\xi < 1.01$. Therefore

$$0 \leq R_1(x) = \frac{3}{2}\sqrt{\xi}(x - 1)^2/2! \leq \frac{3}{2}\sqrt{1.01}(0.1)^2/2 < \frac{3}{2}\left(\frac{1.05}{200}\right) = \frac{3.15}{400}.$$

Part E

- 17.* The given function $f(x)$ defined on $[0, 1]$ to be equal to 1 on the rationals, and equal to 0 on irrationals, is **not** integrable. Indeed, consider the Riemann sums: $S_n = \sum_{i=1}^n f(\xi_i)\Delta x_i$, defined by the partitions of $[0, 1]$ to n equal length subintervals: $0 = x_0 < x_1 < x_2 < \dots < x_n = 1$ with $\Delta x_i = x_i - x_{i-1} = \frac{1}{n}$ where $\xi_i \in [x_{i-1}, x_i]$.

These Riemann sums take the value 1 if ξ_i are all chosen as rational points, and these sums take the value 0 if the ξ_i are all chosen as irrational points. Thus the sums cannot converge, independently of the choice of the ξ_i 's, to a single unique value I as the parameter of partition $1/n$ (=length of the biggest subinterval) tends to 0.

- 17.** Let $f(x) = x$. Prove by means of Darboux sums (that is, upper and lower sums), or Riemann sums, that $f(x)$ is Riemann integrable in the interval $[0, 1]$.

Proof: By means of Darboux sums:

Let $T_n : 0 = x_0 < x_1 < x_2 < \dots < x_n = 1$ be an arbitrary partition of the interval into n subintervals, $\Delta x_i = x_i - x_{i-1}$, and set $\tau = \max_{1 \leq i \leq n} \Delta x_i$ to be the parameter of partition. Let $\xi_i \in [x_{i-1}, x_i]$ be an arbitrary sequence of points. Consider the upper and lower Darboux sums defined by: $\overline{S}_n = \sum_{i=1}^n M_i \Delta x_i$ and $\underline{S}_n = \sum_{i=1}^n m_i \Delta x_i$, where $M_i =$

$\sup\{f(x)|x \in [x_{i-1}, x_i]\} = x_i$, $m_i = \inf\{f(x)|x \in [x_{i-1}, x_i]\} = x_{i-1}$.
Then,

$$0 \leq \overline{S}_n - \underline{S}_n = \sum_{i=1}^n (x_i - x_{i-1}) \Delta x_i = \sum_{i=1}^n (\Delta x_i)^2 \leq \sum_{i=1}^n \tau \Delta x_i = \tau$$

thus as $\tau \rightarrow 0$ the difference between the upper and lower Darboux sums tends to 0, and by a well know theorem on integration this implies the integral exists.